

Activity 1: Heatmap – Student Performance Analysis

```
# Install package if needed
```

```
install.packages("pheatmap")
```

```
library(pheatmap)
```

```
# Data
```

```
marks <- data.frame(
```

```
  Mathematics = c(92,76,81,65,90,58,84,71,95,79),
```

```
  Physics = c(88,72,78,70,94,62,86,75,91,81),
```

```
  Chemistry = c(84,69,83,68,91,60,82,73,93,77),
```

```
  Programming = c(95,80,85,72,96,65,88,77,98,83),
```

```
  English = c(81,74,79,75,89,70,80,72,94,78)
```

```
)
```

```
rownames(marks) <- paste("S",1:10,sep="")
```

```
# Heatmap
```

```
pheatmap(marks,
```

```
  color=colorRampPalette(c("blue","yellow","red"))(50),
```

```
  main="Student Performance Heatmap")
```

```
# Highest Overall Performer
```

```
student_avg <- rowMeans(marks)
```

```
cat("Highest Performer:", names(which.max(student_avg)), "\n")
```

```
# Highest Average Subject
```

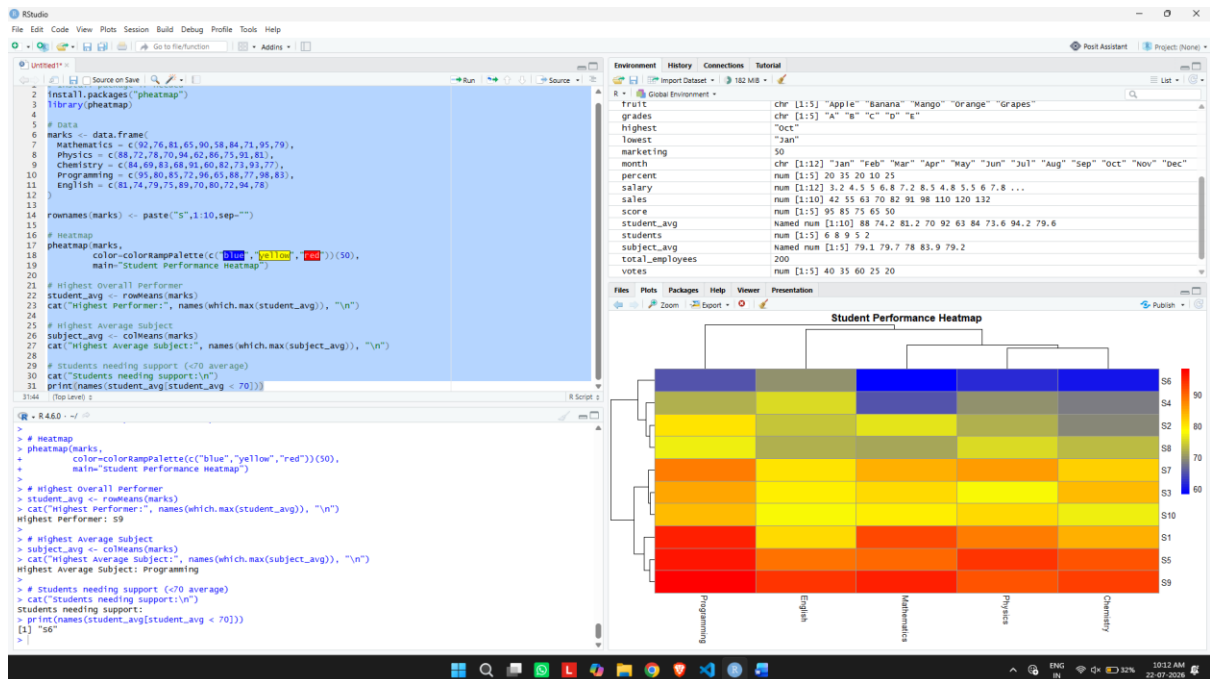
```
subject_avg <- colMeans(marks)
```

```
cat("Highest Average Subject:", names(which.max(subject_avg)), "\n")
```

```
# Students needing support (<70 average)
```

```
cat("Students needing support:\n")
```

```
print(names(student_avg[student_avg < 70]))
```



Activity 2: Violin Plot & Boxplot – Employee Salary

```
install.packages("ggplot2")
```

```
install.packages("reshape2")
```

```
library(ggplot2)
```

```
library(reshape2)
```

```
salary <- data.frame(
```

```
HR=c(4.2,4.5,4.7,4.1,4.6,4.3,4.8,4.4,4.9,4.5),
```

```
IT=c(6.5,7.0,7.4,6.8,8.2,7.6,8.5,7.1,8.8,7.3),
```

```
Finance=c(5.8,6.2,6.0,5.9,6.7,6.4,6.9,6.1,7.2,6.3),
```

```
Marketing=c(5.1,5.3,5.5,5.4,5.8,5.6,5.9,5.2,6.0,5.4)
```

)

```
salary_long <- melt(salary)
```

```
# Violin Plot
```

```
ggplot(salary_long,aes(variable,value,fill=variable))+
```

```
geom_violin()+
```

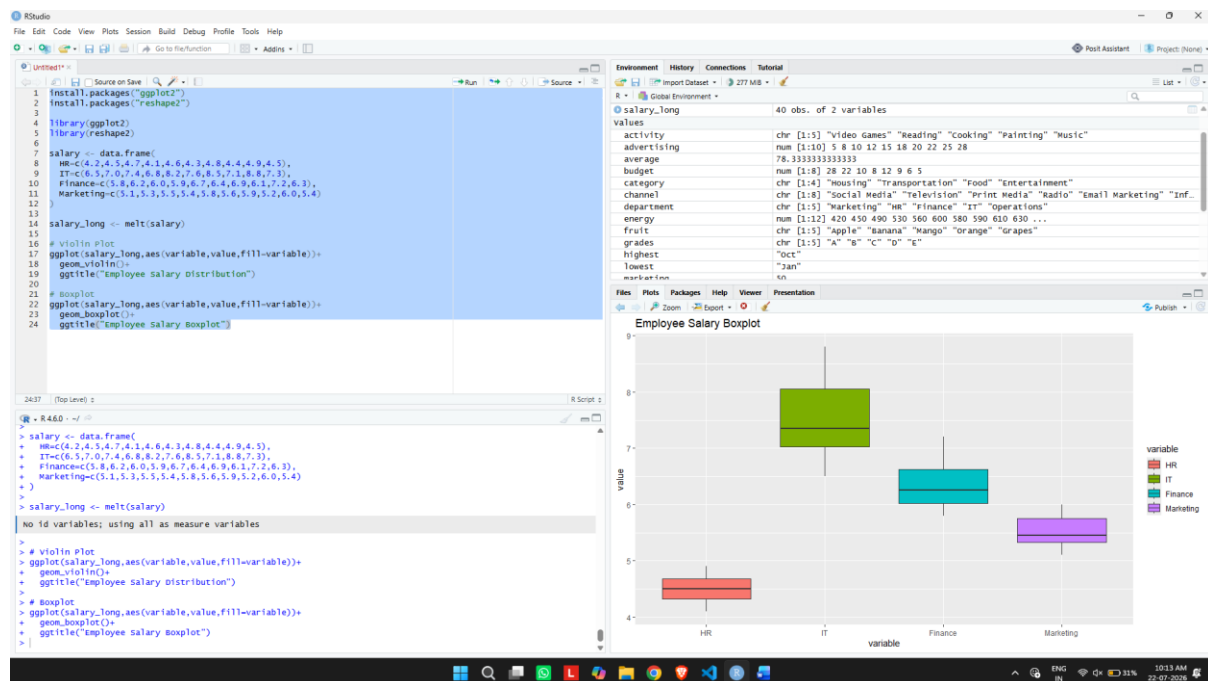
```
ggtitle("Employee Salary Distribution")
```

```
# Boxplot
```

```
ggplot(salary_long,aes(variable,value,fill=variable))+
```

```
geom_boxplot()+
```

```
ggtitle("Employee Salary Boxplot")
```



Activity 3: Histogram & Density Plot – Website Response Time

```
response <- c(210,225,230,240,250,260,270,280,285,290,
```

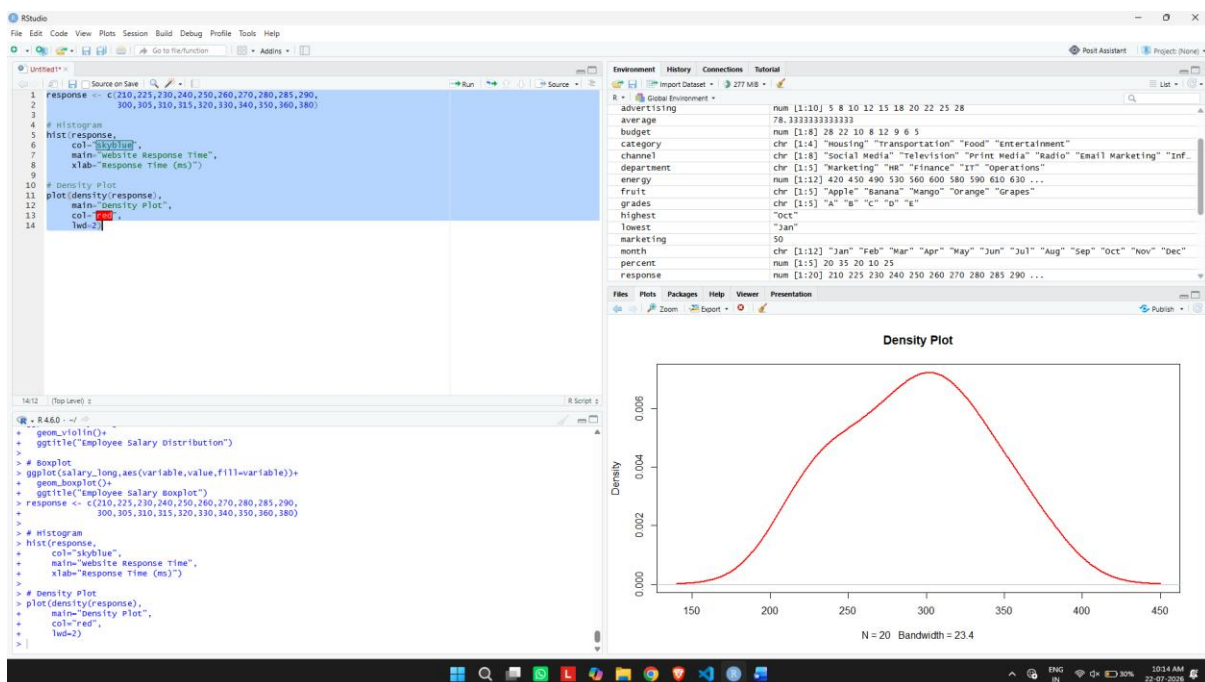
```
300,305,310,315,320,330,340,350,360,380)
```

```
# Histogram
```

```
hist(response,  
  
      col="skyblue",  
  
      main="Website Response Time",  
  
      xlab="Response Time (ms)")
```

```
# Density Plot
```

```
plot(density(response),  
  
      main="Density Plot",  
  
      col="red",  
  
      lwd=2)
```



Activity 4: Ridgeline Plot – Electricity Consumption

```
install.packages("ggribes")
```

```
install.packages("tidyr")
```

```
library(ggribes)
```

```
library(tidyr)
```

```
library(ggplot2)
```

```
electricity <- data.frame(
```

```
  House=paste0("H",1:10),
```

```
  Jan=c(220,240,230,215,225,245,235,228,238,232),
```

```
  Feb=c(235,245,238,228,240,250,242,236,246,239),
```

```
  Mar=c(250,260,255,248,258,265,259,252,262,256),
```

```
  Apr=c(275,285,280,270,282,290,284,278,287,281),
```

```
  May=c(295,305,300,292,298,310,302,297,307,301)
```

```
)
```

```
long <- pivot_longer(electricity,
```

```
  cols=Jan:May,
```

```
  names_to="Month",
```

```
  values_to="Consumption")
```

```
ggplot(long,
```

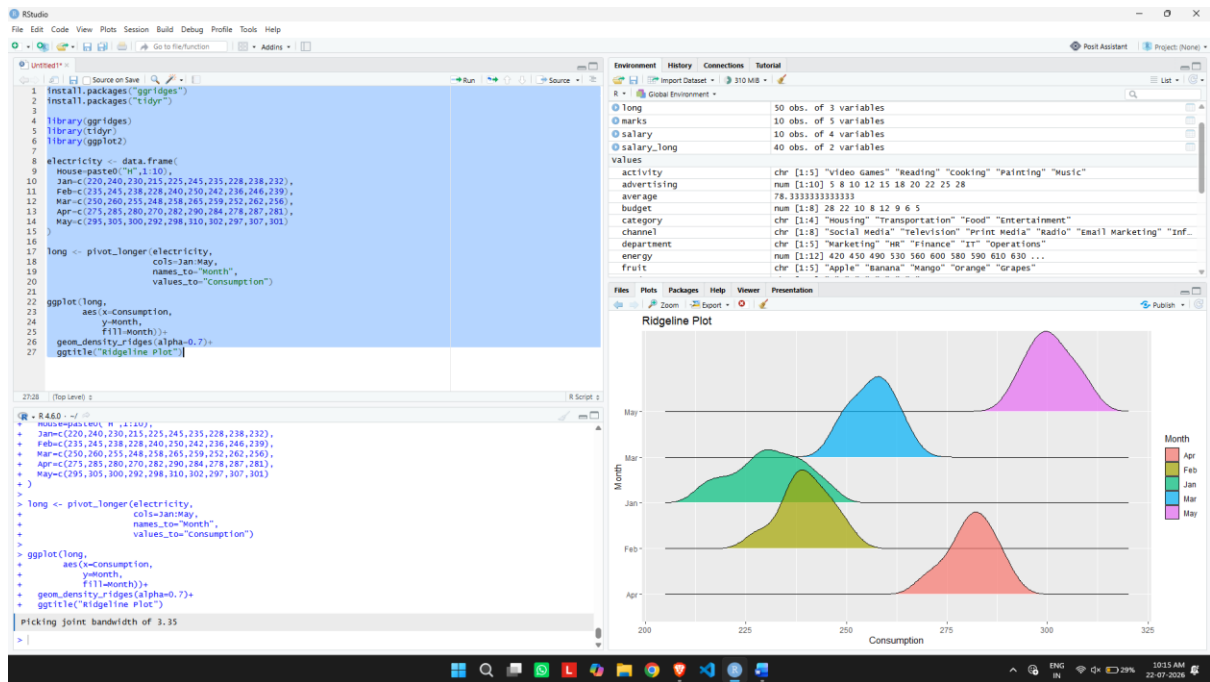
```
  aes(x=Consumption,
```

```
    y=Month,
```

```
    fill=Month))+
```

```
geom_density_ridges(alpha=0.7)+
```

```
ggtitle("Ridgeline Plot")
```



Activity 5: Histogram, Density & Boxplot – Daily Steps

```
steps <- data.frame(
```

```
  Age20_30=c(9500,10200,9800,11000,10500,9700,10100,10800,11200,9900),
```

```
  Age31_45=c(8200,8600,8400,9000,8900,8500,8700,9100,9200,8600),
```

```
  Age46_60=c(6500,6700,6900,7200,7000,6800,7100,7300,7400,6950)
```

```
)
```

```
# Histograms
```

```
par(mfrow=c(1,3))
```

```
hist(steps$Age20_30,col="lightblue",main="20-30")
```

```
hist(steps$Age31_45,col="lightgreen",main="31-45")
```

```
hist(steps$Age46_60,col="pink",main="46-60")
```

```
# Density Curves
```

```
plot(density(steps$Age20_30),
```

```
  col="blue",
```

```

main="Density Curves",

xlab="Steps")

lines(density(steps$Age31_45),col="green")

lines(density(steps$Age46_60),col="red")

legend("topright",

legend=c("20-30","31-45","46-60"),

col=c("blue","green","red"),

lwd=2)

# Boxplot

boxplot(steps,

col=c("skyblue","lightgreen","pink"),

main="Daily Steps by Age Group")

```

